

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes patent literature regarding the synthesis, compounding, and application of Low Acrylonitrile Nitrile Butadiene Rubber (Low Acrylonitrile NBR). The research scope encompasses the polymerization methodologies, structural modifications, and formulation trends of diene-based elastomers containing low levels of acrylonitrile (typically under 30% by mass). Low Acrylonitrile NBR is highly valued for its low-temperature flexibility, oil resistance, and compatibility with polar fillers like silica. The selected patent landscape reveals three primary polymerization and modification approaches: cold emulsion polymerization, anionic solution copolymerization, and olefin metathesis-driven segment interchange. Emulsion polymerization remains the dominant industrial method, utilizing rosin/fatty acid emulsifiers, mercaptan chain transfer agents, and organic peroxide initiators at low temperatures (e.g., 10°C) to achieve random monomer distribution and narrow glass transition widths. Anionic solution polymerization, initiated by functionalized lithium amides, is utilized to introduce tertiary amine or pyrrolidine groups, significantly enhancing filler-elastomer interactions and reducing hysteresis. Olefin metathesis, utilizing Grubbs-type catalysts, offers a route to novel block copolymers via segment interchange of hydrogenated NBR (HNBR). In terms of application, Low Acrylonitrile NBR is heavily featured in electro-conductive matrix-domain structures for electrophotographic members, where its specific volume resistivity and compatibility with epichlorohydrin or other polar rubbers prevent transfer defects and image fogging. This report provides a comprehensive analysis of these synthetic methodologies and compounding formulations to guide the development of high-performance Low Acrylonitrile NBR materials.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3715957B1
- Patent Title: Electro-conductive member, process cartridge image forming apparatus and a ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: EP
- Publication Year: 2022
- Relevance: Mentions matrix-domain structures using rubbers, potentially low acrylonitrile NBR as a component in electro-conductive members.

Polymerization / Synthesis Method - Monomers: Value not specified. - Polymerization Temperature: Value not specified. - Initiator: Value not specified. - Coagulation Method: Value not specified.

Relevant Experimental Evidence - No specific experimental examples disclosed in the supplied evidence.

Technical Relevance Contributes to understanding the application of NBR/rubber blends in electro-conductive layers with matrix-domain structures for electrophotographic members.

Patent 2

- Patent Number: US11340553B2
- Patent Title: Process cartridge and electrophotographic apparatus
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2022
- Relevance: Uses NBR and other rubbers in matrix-domain electro-conductive layers.

Polymerization / Synthesis Method - Solid Content: 23 % (hydroxygallium phthalocyanine pigment) - Solid Content: 88 % - Solid Content: 98 % - Orthophthalonitrile: 5.46 parts - Gallium trichloride: 3.75 parts - Reaction Time: 4.5 hours - Concentrated sulfuric acid: 139.5 parts - Ice: 620 parts - Aqueous ammonia: 2 % - Hydrochloric acid: 200 parts (35 mass %) - Ion-exchanged water: 1000 parts - Chlorogallium phthalocyanine pigment yield: 71 % - Hydroxygallium phthalocyanine pigment yield: 97 % - Stearic acid: 1.33 parts (MRC 9) - Stearic acid: 0.91 parts (MRC 13) - Stearic acid: 1.54 parts (MRC 14) - Zinc oxide: 4.62 parts (MRC 14) - Epichlorohydrin-ethylene oxide-allyl glycidyl ether terpolymer: 90 parts - Liquid NBR: 10 parts

Relevant Experimental Evidence - Preparation 1: Synthesis of chlorogallium phthalocyanine pigment using 5.46 parts orthophthalonitrile, 45 parts alpha-chloronaphthalene, and 3.75 parts gallium trichloride heated to 30°C under nitrogen flow for 4.5 hours, yielding 71%. - Preparation 2: Dissolution of 4.65 parts chlorogallium phthalocyanine in 139.5 parts concentrated sulfuric acid, dropwise addition to 620 parts ice water, dispersion in 2% aqueous ammonia, yielding 97% hydrous hydroxygallium phthalocyanine (23% solid content). - Preparation 4: Mixing 10 parts hydroxygallium phthalocyanine with 200 parts 35% hydrochloric acid, adding to 1000 parts ice-cooled ion-exchanged water to obtain 9 parts chlorogallium phthalocyanine. - MRC Production Examples: Formulation of rubber compositions (SBR, EPDM, Butyl, NBR) with fillers and electroconducting agents, including a mixture of 90 parts epichlorohydrin terpolymer and 10 parts liquid NBR.

Technical Relevance Provides detailed preparation methods for phthalocyanine pigments and rubber formulations containing NBR as a blend component for electro-conductive layers.

Patent 3

- Patent Number: US20230015442A1
- Patent Title: Method for separating and recovering lignin and meltable flowable biolignin ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2023
- Relevance: Mentions lignin nitrile rubber compositions using dry lignin powder with nitrile rubber or acrylonitrile butadiene (NBR) where lignin acts as a nano-filler.

Polymerization / Synthesis Method - Biomass: Value not specified. - Lignin solvent: n-butanol, butyl acetate, butyl ester, furfural - Carrier fluid: Value not specified.

Relevant Experimental Evidence - No specific experimental examples disclosed in the supplied evidence.

Technical Relevance Discusses the integration of lignin with nitrile rubber (acrylonitrile butadiene) to form composites, highlighting the limitations of conventional dry lignin powder as a filler.

Patent 4

- Patent Number: US10869391B2
- Patent Title: Garment-type electronic device and method for producing same
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2020
- Relevance: Uses stretchable conductor compositions containing synthetic rubbers (which can include NBR) for garment-type electronic devices.

Polymerization / Synthesis Method - Curing Temperature: Room temperature - Curing Time: 24 hours - Silicone resin thickness: 5 mm - Stretching: 10 %

Relevant Experimental Evidence - Example 5: Two-component curable silicone resin poured into a screen to a thickness of 5 mm and cured at room temperature over 24 hours, evaluated for electrical conductivity maintenance under 10% stretching. - Example 1: Evaluated level difference at the boundary between the electrode portion and wiring portion, and electrical conductivity maintenance under 10% stretching.

Technical Relevance Demonstrates the formulation of stretchable conductive inks/pastes using synthetic rubbers and resins for flexible electronics.

Patent 5

- Patent Number: US12486386B2
- Patent Title: Process for regenerating a monolithic, macro-structural, inter-penetrating ...
- Assignee: Not disclosed in the available patent text.

- Jurisdiction: US
- Publication Year: 2025
- Relevance: Employs sulfidic metathesis to regenerate vulcanized rubbers, which can include NBR or other diene-based elastomers.

Polymerization / Synthesis Method - Organometallic compound ratio: 0.1 to 5.0 parts per hundred parts vulcanized rubber particles - Reaction Temperature: 100-150 °C - Particle Size: 30 mesh (ground tire rubber particles) - Particle Size: greater than 200 mesh

Relevant Experimental Evidence - No specific experimental examples disclosed in the supplied evidence.

Technical Relevance Discloses a method for preparing modified rubbers via sulfidic metathesis of vulcanized rubber particles using organometallic compounds, useful for recycling diene elastomers.

Patent 6

- Patent Number: US11971683B2
- Patent Title: Electrophotographic electro-conductive member, process cartridge, and ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2024
- Relevance: Uses low-acrylonitrile NBR (such as N215SL, which is a low-acrylonitrile NBR grade) for domain formation in electro-conductive layers.

Polymerization / Synthesis Method - Kneader capacity: 6 L - Kneader filling rate: 70 vol % - Blade rotation speed: 30 rpm - Mixing Time 1: 16 minutes - Mixing Time 2: 18 minutes - Relative Humidity 1: 50 % - Relative Humidity 2: 80 % - Conditioning Time: 48 hours - Acrylic polyol solution: 1000 parts (solid content: 100 parts) - Caprolactone-modified acrylic polyol solution: 100 parts (solid content: 70 mass %) - Drying Temperature: 90 °C - Drying Time: 30 minutes - Second rubber type: BR (150B), IR (IR2200L), NBR (N215SL)

Relevant Experimental Evidence - Example 1: Preparation of Carbon Masterbatch (CMB) for domain formation using a 6 L pressurized kneader at 70 vol% filling rate and 30 rpm blade speed, mixing for 16 and 18 minutes. Second rubber includes NBR (N215SL) with Mooney viscosity of 45. - Example 2: Preparation of a coating solution using 1000 parts acrylic polyol solution (100 parts solid content) and caprolactone-modified acrylic polyol, coated and dried at room temperature for 30 minutes, then at 90°C. - Comparative Examples 8 and 9: Observed matrix-domain structures with 0% of domains satisfying specific structural requirements (B1) and (B2).

Technical Relevance Discloses the use of specific NBR grades (e.g., N215SL, a low-acrylonitrile NBR) in matrix-domain electro-conductive layers to suppress image fogging.

Patent 7

- Patent Number: US8557921B2
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2013
- Relevance: Uses HNBR (hydrogenated nitrile butadiene rubber) in cross-metathesis reactions to prepare block copolymers.

Polymerization / Synthesis Method - HPBD: 0.25 g - HNBR: 0.25 g - Reaction Temperature 1: 95 °C - Reaction Temperature 2: 105 °C - Grubbs II catalyst: 8 mg - Grubbs II catalyst: 9 mg - Fumaryl-modified polycarbonate: 0.25 g - Poly(ethylene-co-octene-co-butadiene): 0.25 g - Poly(ethylene-co-butadiene): 0.10 g - Poly(ethylene-co-octene-co-butadiene): 0.40 g - Tetrahydrofuran: 30 ml - Tetrahydrofuran: 50 ml - Recovered polymer product: 0.45 g - Recovered polymer product: 0.47 g - Recovered polymer product: 0.71 g

Relevant Experimental Evidence - Example 1: A toluene solution of 0.25 g HPBD and 0.25 g HNBR warmed to 95°C, reacted with 8 mg Grubbs II metathesis catalyst, yielding 0.45 g of recovered block copolymer after removing volatiles under reduced pressure. - Example 6: A toluene solution of 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-octene-co-butadiene) warmed to 105°C, reacted with 9 mg Grubbs II catalyst, yielding 0.47 g polymer. - Example 7: A toluene solution of 0.10 g poly(ethylene-co-butadiene) and 0.40 g poly(ethylene-co-octene-co-butadiene) warmed to 105°C, yielding 0.47 g recovered polymer analyzed by CRYSTAF. - Example 10: Metathesis product treated with 50 ml THF overnight to extract non-metathesized components, yielding 0.71 g recovered polymer.

Technical Relevance Demonstrates polymer segment interchange via olefin metathesis using HNBR to synthesize novel block copolymers with tailored Tg and solubility parameters.

Patent 8

- Patent Number: US11137716B2
- Patent Title: Process cartridge and electrophotographic apparatus
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2021
- Relevance: Uses Nipol DN401LL (which is a low-acrylonitrile NBR grade, ~18% AN content) in electro-conductive rubber compositions.

Polymerization / Synthesis Method - Acrylonitrile butadiene rubber (Nipol DN401LL): 70 parts - Epichlorohydrin/ethylene oxide/allyl glycidyl ether terpolymer: 30 parts - Kneader capacity: 6 L - Kneader filling rate: 70 vol % - Blade rotation speed: 30 rpm - Mixing Time 1: 16 minutes - Mixing Time 2: 20 minutes - Butyl rubber (JSR Butyl 065): 100 parts - Calcium carbonate (Nanox #30): 70 parts - Zinc oxide: 7 parts - Zinc stearate: 2.8 parts - CMB 1: 25 parts - MRC 1: 85 parts - Sulfur: 3 parts - Tetramethylthiuram disulfide: 1 part - Epichlorohydrin rubber (EPICHLROMER CG102): 100

parts - Adekacizer LV70: 3 parts - Calcium carbonate: 60 parts - Zinc oxide: 5 parts - Sulfur: 1.8 parts - Carbon black (ASAHI #35G): 40 parts - Zinc stearate: 3.0 parts - Kneader capacity: 7 L - Dibenzothiazyl disulfide: 2.5 parts - Tetraethylthiuram disulfide: 2.0 parts - Sulfur: 3.0 parts - Foaming Time: 15 minutes

Relevant Experimental Evidence - Example: Mixing 70 parts NBR (Nipol DN401LL) and 30 parts epichlorohydrin terpolymer using a 6 L pressure kneader at 70 vol% filling rate and 30 rpm for 16 minutes. - Example: Compounding 100 parts butyl rubber with 70 parts calcium carbonate, 7 parts zinc oxide, and 2.8 parts zinc stearate. - Example: Mixing 25 parts CMB 1 and 85 parts MRC1, followed by adding 3 parts sulfur and 1 part tetramethylthiuram disulfide, mixing for 20 minutes. - Example: Compounding 100 parts epichlorohydrin rubber with 3 parts Adekacizer LV70, 60 parts calcium carbonate, 5 parts zinc oxide, and 1.8 parts sulfur.

Technical Relevance Provides detailed compounding recipes utilizing low-acrylonitrile NBR (Nipol DN401LL) in electro-conductive matrix-domain structures to prevent transfer black spots.

Patent 9

- Patent Number: US11286376B2
- Patent Title: Process for regenerating a monolithic, macro-structural, inter-penetrating ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2022
- Relevance: Relates to sulfidic metathesis of vulcanized rubbers (which can include NBR) to form interpenetrating networks.

Polymerization / Synthesis Method - Organometallic compound ratio: 0.1 to 5.0 parts per hundred parts vulcanized rubber particles - Reaction Temperature: 100-150 °C - Laminate spacing: 10 nm to 5 microns - Anisotropy ratio: 1.1:1 to 3:1

Relevant Experimental Evidence - No specific experimental examples disclosed in the supplied evidence.

Technical Relevance Describes the creation of a rubber-based heterogeneous matrix with an interpenetrating network of elastomers (e.g., SBR, NR, GTR) via sulfidic metathesis.

Patent 10

- Patent Number: US6693160B1
- Patent Title: Functionalized monomers for synthesis of rubbery polymers
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2004

- Relevance: Synthesizes amine-functionalized monomers that can be copolymerized with conjugated diolefins (like butadiene and acrylonitrile) to produce functionalized rubbery polymers (including NBR).

Polymerization / Synthesis Method - Butadiene ratio: 80 % - Styrene ratio: 19 % - PES ratio: 1 % - PES in hexane: 101.4 g (93% purity) - Butadiene in hexane: 20.27 % - Monomer weight: 100 g - n-Butyllithium: 155.9 mmol (2.5% of 6329 mmol) - Monomer conversion: 96 % - DVB conversion: 73 % - DVB conversion: 80 % - Polymerization Time: 100 minutes - Hexane elution time: 5.7 minutes - CSTR retention time 1: 1 hour - CSTR retention time 2: 0.5 hours - Uncured viscosity strain: 15 % - Piperidine: 1 mole - Aqueous sodium hydroxide: 20-35 % - Divinylbenzene: 1030 g (80% active, 824 g pure) - Pyrrolidine: 450 g (528 ml) - TMEDA: 0.48 ml - n-BuLi: 1.5 ml (1.6 M) - TMEDA: 0.52 ml - n-BuLi: 2.0 ml (1.6 M) - Tin coupled PES-styrene-butadiene: 0.25, 0.5, 2.0, 5.0 % PES - Silica coupling agent: 0 to 5 phr - Silica: 55 phr - Processing oil: 10 phr - Zinc oxide: 3 phr - Stearic acid: 2 phr - Antioxidant: 1.5 phr - Sulfenamide accelerator: 1.5 phr - Sulfur: 1.4 phr

Relevant Experimental Evidence - Examples 103-105: Linear polymerization of 1% PES / 19% Styrene / 80% Butadiene. Premix prepared with 101.4 g of 93% PES in hexane added to 20.27% butadiene in hexane. Total monomer conversion reached 96%. - Example 33: Synthesis of functionalized monomer by reacting 1030 g of 80% divinylbenzene (824 g pure) with 450 g dry pyrrolidine under nitrogen, followed by addition of n-butyllithium at 2.5% molar ratio, achieving 73-80% DVB conversion. - Example 14: Preparation of 25/75 SBR containing 0.5 wt% 2-(N-hexamethyleneimino)-methyl 1,3-butadiene, with polymerization times of 90 and 100 minutes. - Example 22: Preparation of high trans 10/90 SBR containing 0.5% pyrrolidine functional groups, showing 74.5% trans-1,4-polybutadiene units and 7.6% polystyrene.

Technical Relevance Discloses synthesis of amine-functionalized monomers and their copolymerization with dienes to produce functionalized elastomers with low hysteresis and high filler affinity.

Patent 11

- Patent Number: US7041761B2
- Patent Title: Functionalized monomers for synthesis of rubbery polymers
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2006
- Relevance: Copolymerization of dienes with functionalized monomers to produce rubbery polymers with low hysteresis.

Polymerization / Synthesis Method - Butadiene ratio: 80 % - Styrene ratio: 19 % - PES ratio: 1 % - PES in hexane: 101.4 g (93% purity) - Butadiene in hexane: 20.27 % - Divinylbenzene: 1030 g (80% active, 824 g pure) - n-Butyllithium: 155.9 mmol - Pinane hydroperoxide: 1.82 g (44% active) - Monomer conversion: 96 % - Shortstop conversion: 65 % - Polymerization Time: 100

minutes - CSTR retention time 1: 1 hour - CSTR retention time 2: 0.5 hours - Sodium naphthalenesulfonate-formaldehyde dispersant: 4.76 g (47.5% active) - Tripotassium phosphate: 7.2 g - Disproportionated rosin acid: 140 g (20% solution) - Unsaturated fatty acids: 532 g (10% solution) - Aqueous potassium hydroxide: 10 % - Styrene: 580 g - PES: 22.5 g (92% purity) - tert-Dodecylmercaptan: 4.5 g - Latex solid content: 24.7 % - Bound styrene: 24.0 % - Bound PES: 0.9 % - Daxad 14B: 33 g (47.5% active) - Potassium chloride: 25.5 g - Disproportionated rosin acid: 767 g - Styrene: 3245 g - PES: 59.8 g (92% purity) - tert-Dodecylmercaptan: 16.0 g - Pinane hydroperoxide: 10 g (55% active) - Isopropylhydroxylamine: 44 g (15% active) - Sulfuric acid: 10 % wt. - PAMS: 20 g (100% purity) - Isopropylhydroxylamine: 8 g (15% active)

Relevant Experimental Evidence - Example 122: Emulsion polymerization of styrene (580 g), butadiene, and PES (22.5 g, 92% purity) using Daxad 14B (4.76 g), disproportionated rosin acid (140 g), and unsaturated fatty acids (532 g). Initiated with pinane hydroperoxide (1.82 g) at 10°C, shortstopped at 65% conversion, yielding a latex with 24.7% solids. Bound styrene was 24.0%, bound PES was 0.9%. - Example 124: Large-scale emulsion polymerization using 3245 g styrene, 59.8 g PES, 16.0 g tert-dodecylmercaptan, and 10 g pinane hydroperoxide (55% active). Shortstopped at 65% conversion with isopropylhydroxylamine, coagulated with 10% sulfuric acid at pH 3-4. - Example 125: Emulsion polymerization of styrene (580 g), butadiene, and PAMS (20 g, 100% purity) using 4.4 g tert-dodecylmercaptan, initiated with pinane hydroperoxide at 10°C, shortstopped at 65% conversion with isopropylhydroxylamine.

Technical Relevance Provides detailed emulsion polymerization recipes for functionalized diene-based rubbers, demonstrating control over conversion, molecular weight, and functional monomer incorporation.

Patent 12

- Patent Number: US5567815A
- Patent Title: Tertiary amine containing anionic initiators used in preparing polymers and ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 1996
- Relevance: Anionic polymerization of dienes using tertiary amine-containing initiators to produce elastomers with reduced hysteresis.

Polymerization / Synthesis Method - 1,3-Butadiene: 500 g - Hexane: 2000 g - n-Butyl lithium: 10 mmols - Polybutadiene molecular weight: 140,700 - Mw/Mn: 1.31 - Cis-1,4 units: 36.5 % - Trans-1,4 units: 52.7 % - Vinyl units: 10.8 %

Relevant Experimental Evidence - Example 1: Polymerization of 500 g 1,3-butadiene in 2000 g hexane initiated with an anionic initiator system (formed by reacting DIPB, a secondary amine, and n-butyl lithium). The recovered polybutadiene had a molecular weight of 140,700, Mw/Mn of 1.31, and a microstructure of 36.5% cis-1,4, 52.7% trans-1,4, and 10.8% vinyl units.

Technical Relevance Discloses a method for preparing tertiary amine-functionalized anionic initiators and using them to synthesize diene elastomers with improved filler interaction and reduced hysteresis.

Patent 13

- Patent Number: US6995224B2
- Patent Title: Functionalized monomers for synthesis of rubbery polymers
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2006
- Relevance: Copolymerization of dienes with functionalized monomers to produce rubbery polymers with low hysteresis.

Polymerization / Synthesis Method - Butadiene ratio: 80 % - Styrene ratio: 19.5 % - Pyr-AMS ratio: 0.5 % - Butadiene in hexane: 20.27 % - PAMS in hexane: 101.4 g (93% purity) - Monomer weight: 100 g - n-Butyllithium: 155.9 mmoles - Monomer conversion: 96 % - DVB conversion: 73 % - DVB conversion: 80 % - Polymerization Time: 100 minutes - CSTR retention time 1: 1 hour - CSTR retention time 2: 0.5 hours - Pyr-AMS consumption: 26 % (at 2 hours) - SiCl₄ residence time: 90 minutes - Sodium hydroxide: 30 % - 2,3-dibromopropane: 0.1 mol - Hexamethyleneimine: 0.4 mol - Pyr-AMS: 44.2 g (100% purity) - 1,3-butadiene in hexane: 19.64 % - Styrene in hexane: 20.27 % - Isopropylalcohol: 10 % - Silica: 55 phr - Processing oil: 10 phr - Zinc oxide: 3 phr - Stearic acid: 2 phr - Antioxidant: 1.5 phr - Sulfenamide accelerator: 1.2 phr - Sulfur: 1.4 phr

Relevant Experimental Evidence - Example 123: Synthesis of a linear 0.5% Pyr-AMS / 19.5% Styrene / 80% Butadiene SBR copolymer. Monomer premix prepared by adding 44.2 g of 100% Pyr-AMS to 36,000 g of 19.64% 1,3-butadiene in hexane. Copolymerized with styrene in hexane and coupled with silicon tetrachloride. - Example: Synthesis of functionalized monomer by reacting 0.1 mol 2,3-dibromopropane with 0.4 mol hexamethyleneimine in 30% aqueous NaOH at 60°C. - Example: Continuous polymerization in two CSTRs with a total residence time of 90 minutes, followed by coupling with SiCl₄, shortstopped with isopropyl alcohol.

Technical Relevance Details the synthesis of pyrrolidine- and amine-functionalized styrene-butadiene-diene terpolymers, demonstrating improved silica and carbon black dispersion.

Patent 14

- Patent Number: US6562929B2
- Patent Title: Conjugated diene-based rubber and method of producing the same, oil extended ...
- Assignee: Not disclosed in the available patent text.
- Jurisdiction: US
- Publication Year: 2003

- **Relevance:** Synthesizes a low-acrylonitrile NBR-like terpolymer containing 9 to 30% by mass of an olefinically unsaturated nitrile monomer (acrylonitrile), 10 to 50% by mass of an aromatic vinyl monomer (styrene), and 20 to 81% by mass of a conjugated diene monomer (butadiene).

Polymerization / Synthesis Method - Olefinically unsaturated nitrile monomer (acrylonitrile): 9 to 30 % by mass - Aromatic vinyl monomer (styrene): 10 to 50 % by mass - Conjugated diene monomer (butadiene): 20 to 81 % by mass - Mooney viscosity [ML1+4 (100°C)]: 20 to 200 - Glass transition point: -60 to 0 °C - Glass transition width: 20 °C or less - Extending oil: 10 to 60 parts by mass per 100 parts rubber

Relevant Experimental Evidence - Example 1: Copolymerization of monomers containing 9-30% acrylonitrile, 10-50% styrene, and 20-81% butadiene. Acrylonitrile is added portionwise to the polymerization system containing styrene and butadiene, and the remainder is added intermittently to maintain random distribution and a narrow glass transition width (20°C or less). Mooney viscosity of the resulting rubber was in the range of 20 to 200.

Technical Relevance Directly addresses the synthesis of low-acrylonitrile NBR terpolymers with highly randomized monomer distribution, narrow glass transition width, and excellent compatibility with silica fillers.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- **Monomer Content Ranges:** US6562929B2 specifies a highly randomized terpolymer containing 9 to 30% by mass of acrylonitrile, 10 to 50% styrene, and 20 to 81% butadiene. US11137716B2 utilizes Nipol DN401LL, a commercial low-acrylonitrile NBR grade (~18% acrylonitrile), while US11971683B2 utilizes N215SL, another low-acrylonitrile NBR grade.
- **Polymerization Processes:** Emulsion polymerization is detailed in US7041761B2 for functionalized diene rubbers. Anionic solution polymerization is utilized in US5567815A, US6693160B1, and US6995224B2 to prepare functionalized diene elastomers. US8557921B2 introduces a distinct metathesis-driven segment interchange process in solution to form block copolymers from HNBR.
- **Emulsifier and Initiator Systems:** US7041761B2 details a cold emulsion system (10°C) using disproportionated rosin acid and unsaturated fatty acid potassium salts as emulsifiers, initiated by pinane hydroperoxide. In contrast, the anionic solution systems (US5567815A, US6693160B1) utilize n-butyllithium or functionalized lithium amides in hydrocarbon solvents at elevated temperatures.
- **Chain-Transfer and Coagulation:** US7041761B2 utilizes tert-dodecylmercaptan as a chain-transfer agent to control molecular weight and coagulates the latex using 10% sulfuric acid at pH 3-4. The solution polymerization patents typically terminate reactions with alcohols (e.g., isopropyl alcohol) and recover the polymer by vacuum stripping or solvent precipitation.

- Compounding and Applications: US11137716B2 and US11971683B2 focus on compounding low-acrylonitrile NBR into matrix-domain electro-conductive layers with epichlorohydrin rubbers, utilizing sulfur vulcanization systems with zinc oxide and stearic acid to control electrical impedance and physical properties.

4. CONCLUSION

The synthesis of high-performance Low Acrylonitrile NBR requires precise control over monomer randomness, molecular weight distribution, and functionalization. Based on the patent evidence, cold emulsion polymerization at 10°C using disproportionated rosin acid emulsifiers and tert-dodecylmercaptan chain transfer agents is recommended for producing highly randomized terpolymers with narrow glass transition widths (<20°C) and Mooney viscosities of 20-200. For applications requiring low hysteresis and high filler interaction, anionic solution copolymerization with amine-functionalized monomers (such as PES or PAMS) initiated by functionalized lithium amides is highly effective. Compounding low-acrylonitrile grades (such as Nipol DN401LL or N215SL) into matrix-domain structures with epichlorohydrin rubbers provides optimal electro-conductive properties for specialized engineering applications. Future research should focus on optimizing metathesis-driven segment interchange to create thermoplastic NBR elastomers with tailored block architectures.

5. REFERENCES

- EP3715957B1 | Electro-conductive member, process cartridge image forming apparatus and a ... | Not disclosed in the available patent text. | EP | 2022 | <https://patents.google.com/patent/EP3715957B1/en>
- US11340553B2 | Process cartridge and electrophotographic apparatus | Not disclosed in the available patent text. | US | 2022 | <https://patents.google.com/patent/US11340553B2/en>
- US20230015442A1 | Method for separating and recovering lignin and meltable flowable biolignin ... | Not disclosed in the available patent text. | US | 2023 | <https://patents.google.com/patent/US20230015442A1/en>
- US10869391B2 | Garment-type electronic device and method for producing same | Not disclosed in the available patent text. | US | 2020 | <https://patents.google.com/patent/US10869391B2/en>
- US12486386B2 | Process for regenerating a monolithic, macro-structural, inter-penetrating ... | Not disclosed in the available patent text. | US | 2025 | <https://patents.google.com/patent/US12486386B2/en>
- US11971683B2 | Electrophotographic electro-conductive member, process cartridge, and ... | Not disclosed in the available patent text. | US | 2024 | <https://patents.google.com/patent/US11971683B2/en>
- US8557921B2 | Production of meta-block copolymers by polymer segment interchange | Not disclosed in the available patent text. | US | 2013 |

<https://patents.google.com/patent/US8557921B2/en>

- US11137716B2 | Process cartridge and electrophotographic apparatus | Not disclosed in the available patent text. | US | 2021 | <https://patents.google.com/patent/US11137716B2/en>
- US11286376B2 | Process for regenerating a monolithic, macro-structural, inter-penetrating ... | Not disclosed in the available patent text. | US | 2022 | <https://patents.google.com/patent/US11286376B2/en>
- US6693160B1 | Functionalized monomers for synthesis of rubbery polymers | Not disclosed in the available patent text. | US | 2004 | <https://patents.google.com/patent/US6693160B1/en>
- US7041761B2 | Functionalized monomers for synthesis of rubbery polymers | Not disclosed in the available patent text. | US | 2006 | <https://patents.google.com/patent/US7041761B2/en>
- US5567815A | Tertiary amine containing anionic initiators used in preparing polymers and ... | Not disclosed in the available patent text. | US | 1996 | <https://patents.google.com/patent/US5567815A/en>
- US6995224B2 | Functionalized monomers for synthesis of rubbery polymers | Not disclosed in the available patent text. | US | 2006 | <https://patents.google.com/patent/US6995224B2/en>
- US6562929B2 | Conjugated diene-based rubber and method of producing the same, oil extended ... | Not disclosed in the available patent text. | US | 2003 | <https://patents.google.com/patent/US6562929B2/en>